Assignment 3

Note: You are advised to use modular programming approach with consideration of basic functionalities of JAVA (not build up libraries). To initialize the object always use constructor.

1. You are required to create a Java class to maintain your personal record, which includes your name, father's name, date of birth, and age (computed with respect to the current date). The age should be displayed in years, months, and days. Display the record on the screen such that each field starts on a new line. For example, the student's name should be displayed as "Name of the student:" followed by its corresponding value. Use the Scanner class to accept input from the keyboard. Initialize the instance variables using both a constructor and a block constructor (instance initialization block). Follow a class-and-object-based modular programming approach.
2. You are required to create a Java class to compute the perimeter, area, and diagonal of a rectangle. The length and breadth should be accepted from the keyboard as float and double data types, respectively, and initialized using a parameterized constructor. Use the concept of method overloading to implement the methods for computing the perimeter, area, and diagonal of the rectangle. Follow a class-and-object-based modular programming approach.
3. You are required to create a Java class to compute the Greatest Common Divisor (GCD) and Least Common Multiple (LCM) of five integers. Store the five integers as static data members and initialize them using a static initialization block. Implement method overloading to compute the GCD and LCM for different numbers of arguments (for example, two integers, three integers, and five integers).
4. You are required to design a class for the student whose different entries are Name, Registration ID, Father’s Name, Degree Name, Current Semester, and CPI. While entering these data students have given flexibility to use either of numeric and string for Registration ID and current semester. You are required to compute number of students in particular semester, number of students having CPI below 6, number of students enrolled for particular degree program, number of students opted registration ID as string and number of students opted current semester as numeric. Write the program for 100 students using generic class concept along with utility class for the computation required use it in generic class for actual computation.
5. Write a program to create smart environment monitoring system which has multiple sensors having different data format collected from environment. These sensors include Temperature sensor, Humidity sensor, PPM sensor, AQI sensor and many more. Sensors have some common data along with individual one. Write the program which have least redundancy and use utility and generic concepts of reutilizations. Create a sensor class and use constructor to initialize every different type of sensor and use constructor chaining for initializing the same data used by every sensor.